Bosch-Age and Gender detection

Team 12

Inter IIT Tech Meet 10.0

March 24, 2022

Overview

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- Assumption
- Intuition
- Our Approach
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Objective

- From a surveillance video detect all people.
- For the people detected run, an algorithm to predict age and gender of the said person.

Assumptions

- Video is from a certain oblique angle
- The frame rate of the video is around 15 fps
- The camera used for the video is static so the optical flow of objects is relatively noise free
- The quality of video is assumed to be atleast 720p.
- The video is in RGB colour channels and not others(B&W for example)

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- Most face detection models perform poorly when the object is far away and the video is taken at an angle therefore we need some robust model which can atleast detect people correctly.
- For people whose face was not identified by the face detection model, we check if there are alternative model that can predict some information about gender/age with reasonable accuracy.

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- If a face is detected we use age_net and gender_net to predict age and gender prediction.
- If a face is not detected we pass this to a model which predicts gender based on the body detected.

Models used

- YOLOv5s[1]: For detecting people in video frames.
- MTCNN[2]: For detecting faces in the people detected.
- age_net[3]: For predicting age on faces detected.
- gender_net[4]: For predicting gender on the faces detected.
- body2gen : For predicting gender on people whose faces could not be detected.

[1]YOLO : You Only Look Once
[2]MTCNN
[3]age_net : predicting age from faces
[4]gender_net : predicting gender from faces

Model Details

- YOLO : Version 5 pretrained model used from torch.hub
- MTCNN : Used from mtcnn library of python
- age_net : Publicly available caffemodel and prototype
- gender_net : Publicly available caffemodel and prototype
- body2gen : Fine tuned after using VGG-19 base and configuring it to predict gender based on bodies detected.

Workflow

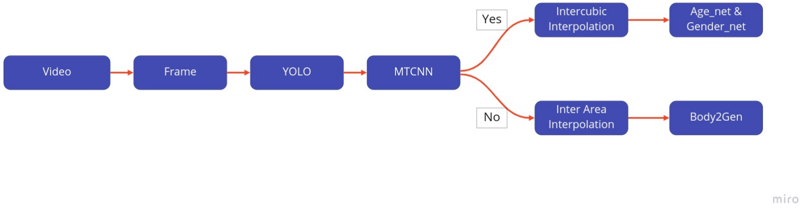

Figure: Workflow of our model

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- Our model is adaptive to a large number of people in a frame, and frame sampling rate can be adjusted as a parameter based on latency and accuracy requirements.

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[6] PETA : Pedestrian dataset

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- For people whose faces are not detected by MTCNN, we have trained a VGG-19[5]derivative model body2gen which predicts gender based on just the body, trained on the PETA dataset[6].
- Our model is resilient to changes in frame rate, quality of the video, and even works reasonably in a crowded environment.

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- From PETA dataset it is possible to train a model which can predict age from just the body thus the need to detect faces can be circumvented entirely, solving the problem of relying on detecting faces entirely.
- Given that we are likely to encounter side-views of faces of people we can look for ways to integrate models (GANs) that generate front views of faces from side-views into the pipeline for more robust prediction.